The Normal Distribution

Chapter Overview

1: Characteristics of the Normal Distribution

What shape is it? What parameters does it have?

2: Finding probabilities on a standard normal curve.

"Given that IQ is distributed as $X \sim N\left(100, 15^2\right)$, determine the probability that a randomly chosen person has an IQ above 130."

3: Finding unknown means/standard deviations.

In Wales, 30% of people have a height above 1.6m. Given the mean height is 1.4m and heights are normally distributed, determine the standard deviation of heights.

4: Binomial $\rightarrow$ Normal Approximations

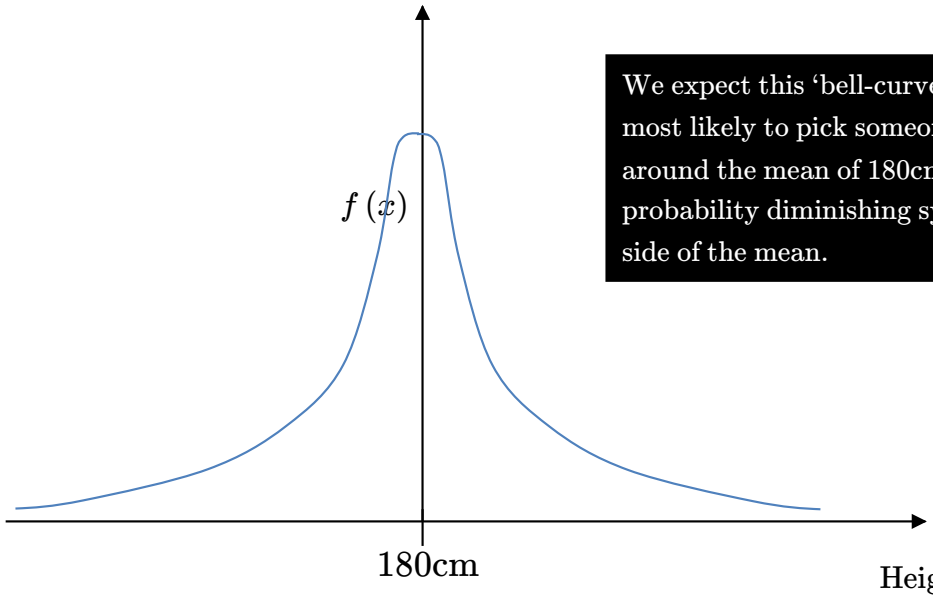
How would I approximate $X \sim B(10, 0.4)$ using a Normal distribution? Under what conditions can we make such an approximation?

5: Hypothesis Testing

What does it look like?

Solution 3

The following shows what the probability distribution might look like for a random variable X, if X is the height of a randomly chosen person.



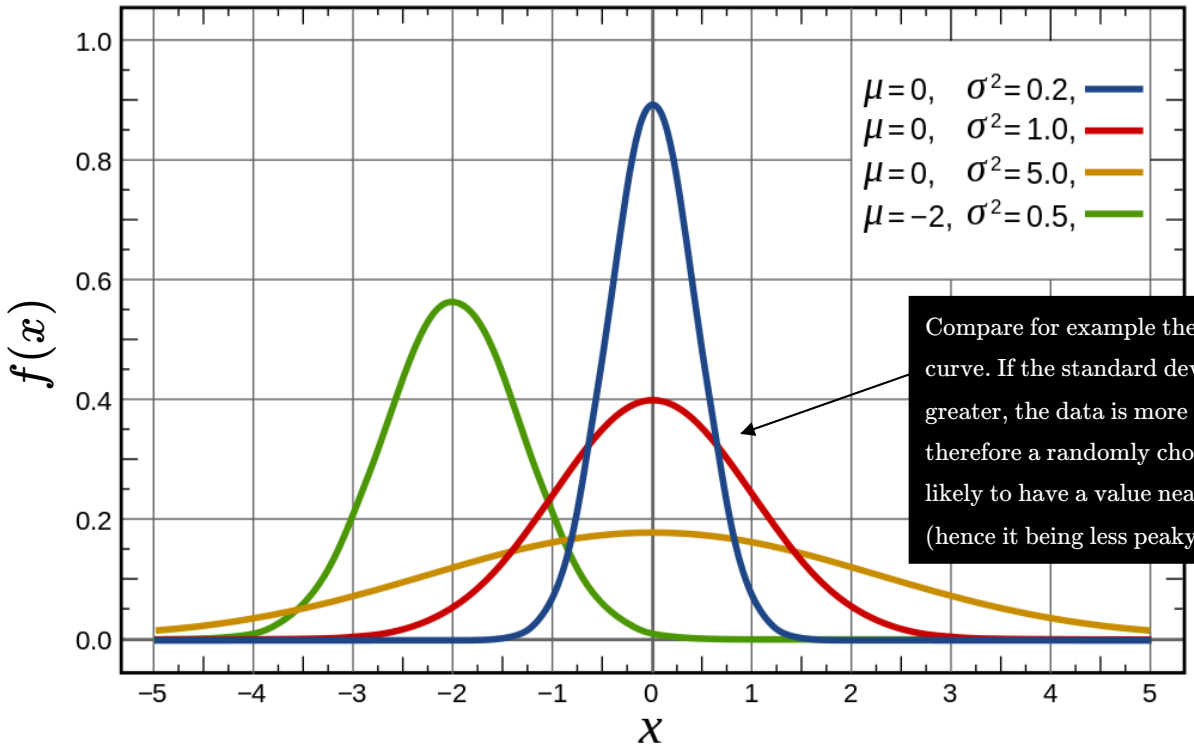
We expect this 'bell-curve' shape, where we're most likely to pick someone with a height around the mean of 180cm, with the probability diminishing symmetrically either side of the mean.

A variable with this kind of distribution is said to have a **normal distribution**.

For normal distributions we tend to draw the y axis at the mean for symmetry.

What does it look like?

Solution 4



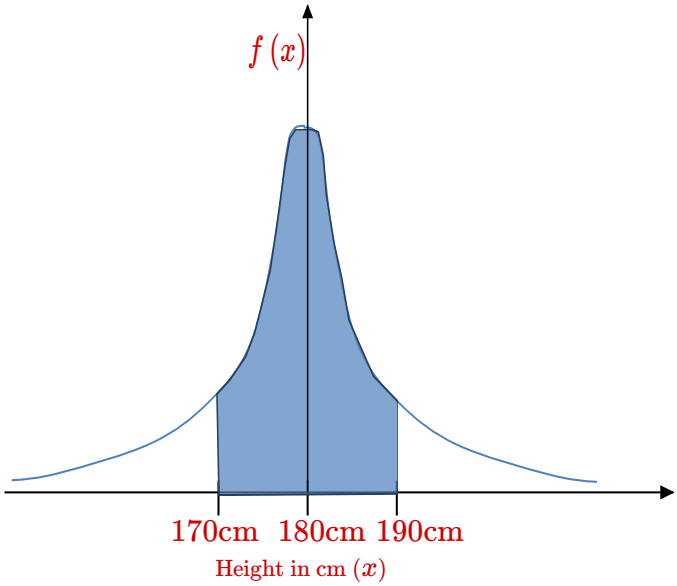
Compare for example the blue and red curve. If the standard deviation is greater, the data is more spread out, and therefore a randomly chosen thing is less likely to have a value near the mean (hence it being less peaky).

We can set the mean μ and the standard deviation σ of the Normal Distribution. If a random variable X is normally distributed, then we write

$$\mathbf{X \sim N\left(\mu, \sigma^2\right)}$$

Normal Distribution Q & A

Solution 5



Q1. What kind of variable can be modelled by a normal distribution?

Q1 A continuous variable.

Q2. Discrete probabilities sum to 1. What condition holds for a continuous distribution?

Q2 The total area under the density curve is 1.

Q3. How do we find $P(170 < X < 190)$?

Q3 Find the area under the curve between 170 and 190.

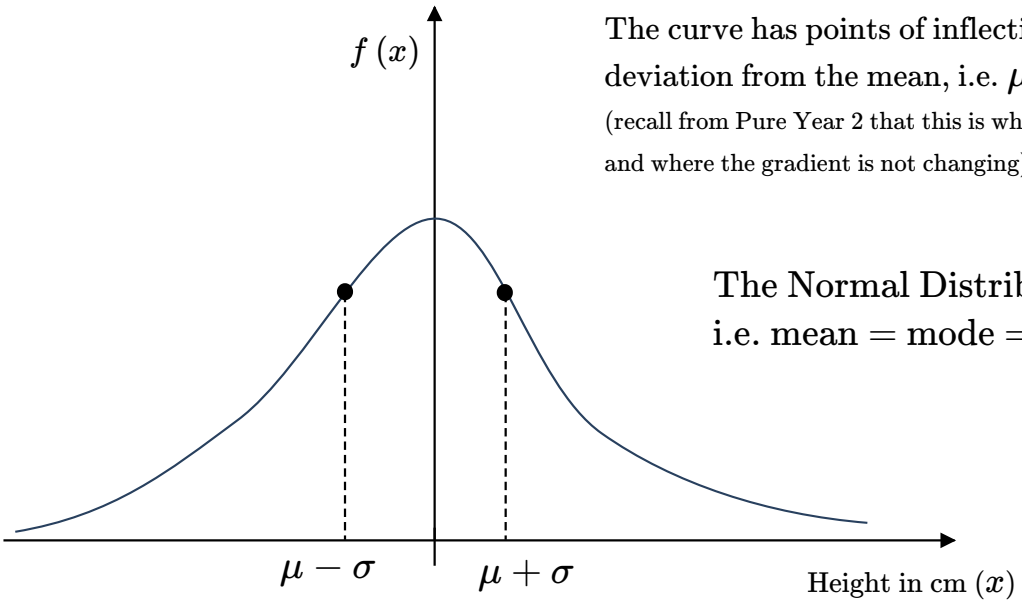
Q4. What is the probability that height is exactly 200 cm?

Q4 $P(X = 200) = 0$. A single point has zero area under a continuous density curve.

The vertical axis gives probability density per centimetre. As in a histogram, area represents probability. A continuous variable has a density function $f(x)$.

Further Facts

Solution 6



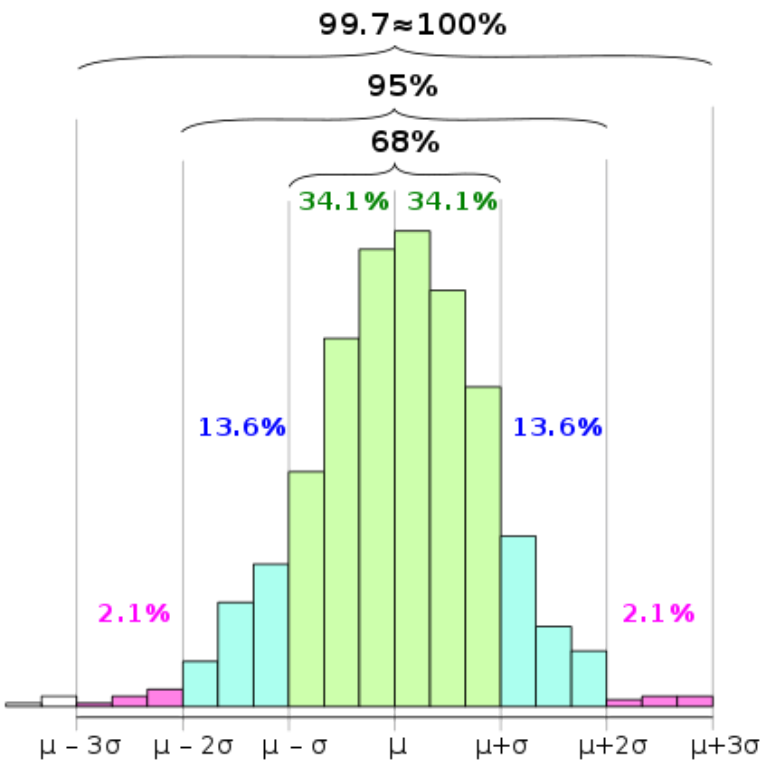
The curve has points of inflection one standard deviation from the mean, i.e. $\mu \pm \sigma$

(recall from Pure Year 2 that this is where the curve changes concavity, and where the gradient is not changing)

The Normal Distribution is symmetrical, i.e. mean = mode = median

The 68-95-99.7 rule

Solution 7



The histogram above is for a quantity which is approximately normally distributed.

The 68-95-99.7 rule is a shorthand used to remember the percentage of data that is within 1, 2 and 3 standard deviations from the mean respectively.

You need to memorise this!

$\approx 68\%$ of data is within one standard deviation of the mean.
$\approx 95\%$ of data is within two standard deviations of the mean.
$\approx 99.7\%$ of data is within three standard deviations of the mean.

For practical purposes we consider all data to lie within $\mu \pm 5\sigma$

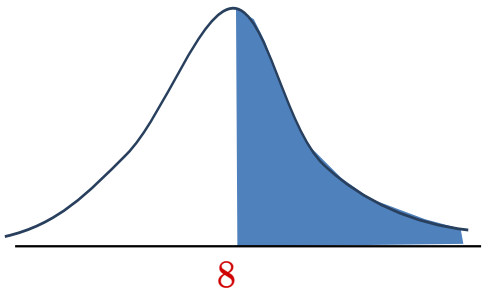
Examples

Solution 8

[Textbook] The diameters of a rivet produced by a particular machine, X mm, is modelled as $X \sim N\left(8, 0.2^2\right)$. Find:
a) $P\left(X > 8\right)$
b) $P\left(7.8 < X < 8.2\right)$

Tip: Draw a diagram!

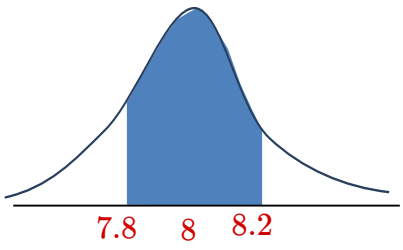
a



8 is the mean, so by the symmetry of the normal distribution, 50% of the area lies above the mean.

$\therefore P\left(X > 8\right) = 0.5$

b



The standard deviation is 0.2, so the data lies within $\mu \pm \sigma$

$\therefore P\left(7.8 < X < 8.2\right) = 0.68$

Test Your Understanding

Solution 9

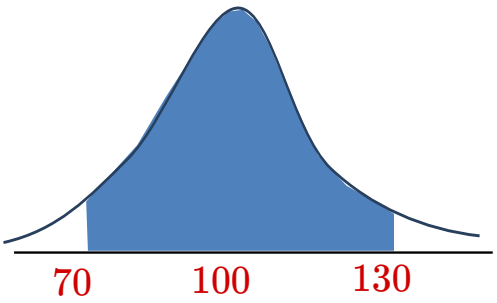
IQ ("Intelligence Quotient") for a given population is, by definition, distributed using $X \sim N\left(100, 15^2\right)$. Find:

a) $P\left(70 < X < 130\right)$

b) $P\left(X > 115\right)$

Tip: Draw a diagram!

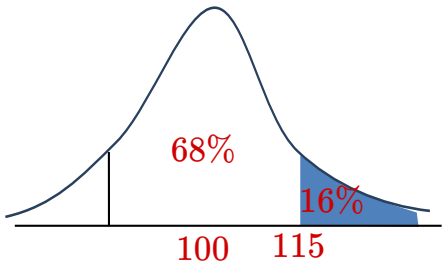
a



We know that 95% of data lies within 2 standard deviations of the mean.

$\therefore P\left(70 < X < 130\right) = 0.95$

b



68% lies within one standard deviation, so there must be 16% at each tail.

$\therefore P\left(X > 115\right) = 0.16$

Exercise 3A – Statistics Y2 pages 40-41

Solution 10

6. Dormouse masses D grams follow $N(\mu, 25)$. If 97.5% weigh less than 70 g, find μ.

7. Pig masses M kg follow $N(\mu, \sigma^2)$. If 84% weigh more than 52 kg and 97.5% weigh more than 47.5 kg, find μ and σ^2.

a

Using the approximate 95% rule, 70 g is about two standard deviations above the mean. Since $\sigma = \sqrt{25} = 5$ g, $\mu \approx 70 - 2(5) = 60$ g.

b

Using the 68% and 95% rules, about 16% lie below $\mu - \sigma$ and 2.5% below $\mu - 2\sigma$.

$52 \approx \mu - \sigma, 47.5 \approx \mu - 2\sigma$.

Subtracting gives $\sigma \approx 4.5$ kg, so $\sigma^2 \approx 20.25$ kg^2 and $\mu \approx 56.5$ kg.

Exercise 3A – Statistics Y2 pages 40-41

Solution 11

8. Test scores S (percentages) follow a normal model with mean 45 and standard deviation 15. Find:

(a) $P(S > 45)$

(b) $P(30 < S < 60)$

(c) $P(15 < S < 75)$

(d) Alexia says this model is unsuitable since scores above 100% are impossible. Is she correct? Give a reason.

a

By symmetry, half the scores are above 45, so the probability is 0.5.

b

Within one standard deviation of the mean: approximately 0.68.

c

Within two standard deviations: approximately 0.95.

d

100 is more than three standard deviations above the mean. The model gives a very small probability to impossible scores above 100, so it can still be a reasonable approximation.

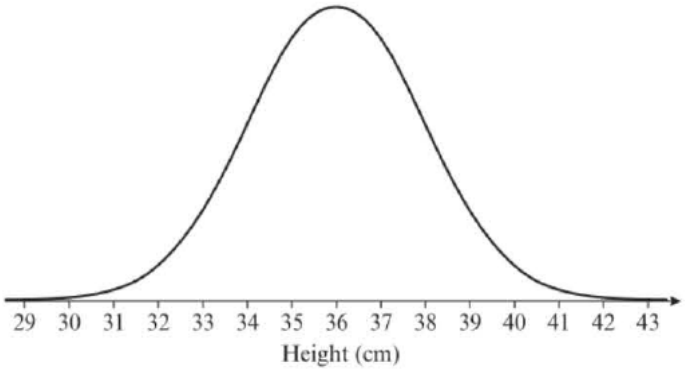
Exercise 3A – Statistics Y2 pages 40-41

Solution 12

9. The diagram shows UK barn owl heights in cm. An ornithologist models them using a normal distribution.

(a) State the mean height.

(b) Estimate the standard deviation.



a

The peak is at approximately 36 cm.

b

The points of inflection are at $\mu \pm \sigma$. They appear between 33 and 34 cm and between 38 and 39 cm. Hence σ is about 2–3 cm.

Getting normal values from your calculator — Solution 13

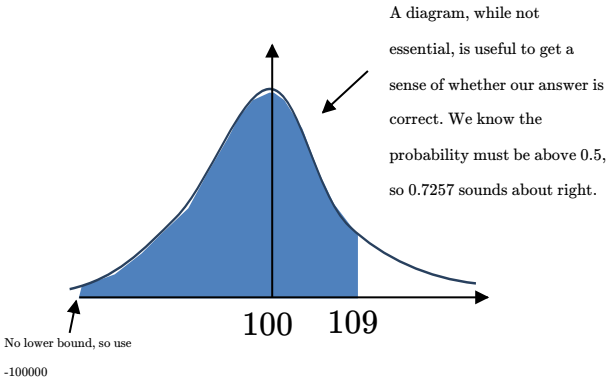
https://education.casio.co.uk/resources/videos/normal-distribution-probabilities-and-values

Just like a cumulative frequency graph gives the running total of the frequency up to a given value, a **cumulative distribution** gives the **running total of the probability** up to a given value.

IQ is distributed using $X \sim N\left(100, 15^2\right)$. Find
(a) $P\left(X < 109\right)$
(b) $P\left(X \geq 93\right)$
(c) $P\left(110 < X < 120\right)$
(d) $P\left(X < 80 or X > 106\right)$

a

1. Press MODE. Stats
2. Choose DISTRIBUTION
3. Choose Normal CD (i.e. "Cumulative Distribution")
4. Since the lower value is effectively $-\infty$, use any value at least 5σ below the mean (-100000 will do!). Press = after each value.
5. Put the upper value as 109.
6. Set $\sigma = 15$ and $\mu = 100$
7. You should obtain $P\left(X < 109\right) = 0.7257(4dp)$



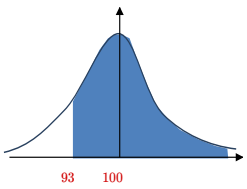
A diagram, while not essential, is useful to get a sense of whether our answer is correct. We know the probability must be above 0.5, so 0.7257 sounds about right.

Getting normal values from your calculator — Solution 14

IQ is modelled by $X \sim N(100, 15^2)$. Find:
(a) $P(X < 109)$
(b) $P(X \geq 93)$
(c) $P(110 < X < 120)$
(d) $P(X < 80 \text{ or } X > 106)$.

b

b Use 93 for the lower limit and something arbitrarily high (e.g. 100000) for the upper limit.
P(X ≥ 93) = 0.6796 (4 dp)



Note: ≥ vs > makes no difference as the distribution is continuous.

c

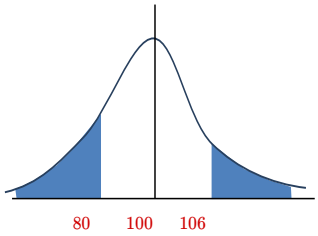
c $P\left(110 < X < 120\right) = 0.1613(4dp)$

d

d

From the graph, we can see that we can find the probability between 80 and 106, and take away from 1:

$P\left(X < 80 or X > 106\right) = 1 - P\left(80 < X < 106\right)$
$= 1 - 0.5642$
$= 0.4358$



Tip. Return to normal calculation mode to subtract from 1. ANS stores the last distribution result.

Test Your Understanding

Assume IQ $X \sim N(100, 15^2)$. Mensa requires an IQ of at least 131.

(a) What percentage of the population is eligible?

(b) Of 30 randomly chosen adults, what is the probability that at least three are eligible? Hint: use the binomial distribution.

Solution 15

a

$P(X \geq 131) = 0.01938$, so approximately 1.938% are eligible.

b

Let Y count eligible adults. $Y \sim B(30, 0.0193827)$.

$P(Y \geq 3) = 1 - P(Y \leq 2)$

$= 1 - 0.979986 = 0.02001$ (4 s.f.).

Use the calculator's binomial cumulative probability function.

Exercise 3B – Statistics Y2 pages 40-41

8. Soap volume per press, X ml, follows $N(6, 0.8^2)$.

(a) Find (i) $P(X > 7)$ and (ii) $P(X < 5)$.

(b) Find the probability that all three presses dispense less than 5 ml.

Solution 16

8a(i)

$P(Z > 1.25) = 0.1056$.

8a(ii)

$P(Z < -1.25) = 0.1056$.

8b

Independent presses give $0.1056498^3 \approx 0.0012$.

9. Bottled mineral water volume W ml follows $N(500, 14^2)$.

(a) Find (i) $P(W > 505)$ and (ii) $P(W < 490)$.

(b) Find the probability all four sampled bottles contain more than 490 ml.

9a(i)

$P(Z > 5/14) = 0.3605$.

9a(ii)

$P(Z < -10/14) = 0.2375$.

9b

$P(W > 490) = 1 - 0.23752\ldots$; by independence, $0.76248\ldots^4 = 0.3380$.

Exercise 3B – Statistics Y2 pages 40-41

Solution 17

11. Bolt diameters D mm follow $N(13, 0.1^2)$.

(a) Find the probability a bolt is smaller than 12.8 mm. (1)

(b) A perfect bolt has diameter between 12.9 and 13.1 mm. In a random sample of 40, find the probability of more than 25 perfect bolts. (4)

a

$P(D < 12.8) = \Phi(-2) = 0.0228$ (4 d.p.).

b

$p = P(12.9 < D < 13.1) = 0.68269\ldots$.
With $Y \sim B(40, p)$, $P(Y > 25) = 1 - P(Y \leq 25) \approx 0.7345$.

Exercise 3B – Statistics Y2 pages 40-41

Solution 18

12. Squirrel masses X g follow $N(480, 40^2)$.

(a) Find the probability a squirrel has mass above 490 g. (1)

(b) A random sample of 30 is taken. Find the probability at least 15 have masses between 470 g and 490 g. (4)

a

$P(X > 490) = 1 - \Phi(0.25) = 0.4013$.

b

$p = P(470 < X < 490) = 0.19741\ldots$.
With $Y \sim B(30, p)$, $P(Y \geq 15) = 1 - P(Y \leq 14) \approx 0.0001986$.

Inverse Normal Distribution

Solution 19

We now know how to use a calculator to value of the variable to obtain a probability. But we might want to do the reverse: given a probability of being in a region, how do we find the value of the boundary?

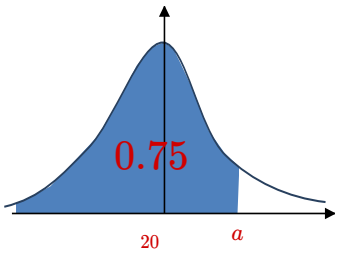
[Textbook] $X \sim N\left(20, 3^2\right)$. Find, correct to two decimal places, the values of a such that:

a. $P\left(X < a\right) = 0.75$

b. $P\left(X > a\right) = 0.4$

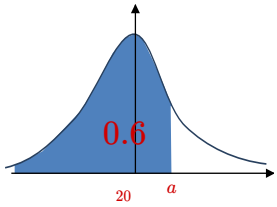
c. $P\left(16 < X < a\right) = 0.3$

a Choose Inverse Normal. Enter lower-tail area 0.75, mean 20 and standard deviation 3.
a = 22.0235… ≈ 22.02.

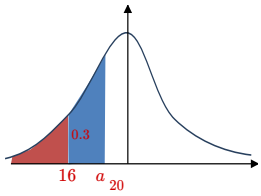


The "area" requested by your calculator is the probability up to the value of interest (in this case a).

DRAW A SKETCH!



b P(X > a) = 0.4
So P(X < a) = 0.6
(i.e. if 40% is above a, then 60% must be below)
a = 20.76



c Find the area up to 16, then add 0.3 to find the area up to a:
P(X < 16) = 0.09121
P(X < a) = 0.39121
So a = 19.17

REALLY, DRAW A SKETCH!

Further Example

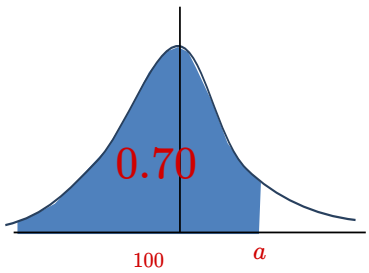
Solution 20

If the IQ of a population is distributed using $X \sim N\left(100, 15^2\right)$.

a. Determine the IQ corresponding to the top 30% of the population.

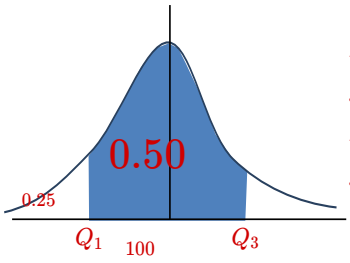
b. Determine the interquartile range of IQs.

a



$P\left(X > k\right) = 0.3$
$\therefore P\left(X < k\right) = 0.7$
$k = 107.87 (2dp)$

b



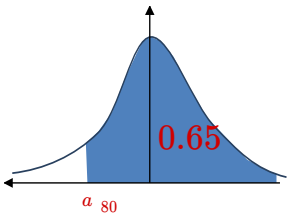
$P\left(X < Q_1\right) = 0.25$
$\therefore Q_1 = 89.88$
$P\left(X < Q_3\right) = 0.75$
$\therefore Q_3 = 110.12$
$IQR = 110.12 - 89.88 = 20.24$

In general the quartiles of a normal distribution are approximately $\mu \pm \frac{2}{3}\sigma$

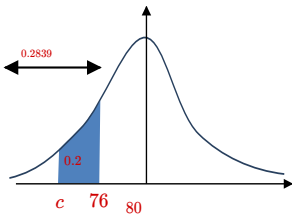
Test Your Understanding

Solution 21

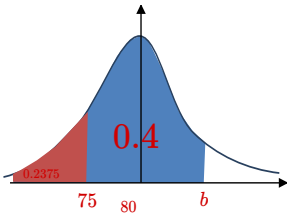
$X \sim N\left(80, 7^2\right)$. Using your calculator,
a. determine the a such that $P\left(X > a\right) = 0.65$
b. determine the b such that $P\left(75 < X < b\right) = 0.4$
c. determine the c such that $P\left(c < X < 76\right) = 0.2$
d. determine the interquartile range of X.



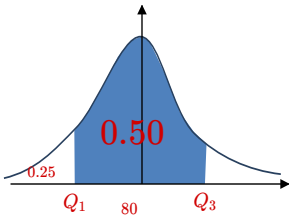
a P(X > a) = 0.65
So P(X < a) = 0.35
a = 77.303 (3 dp)



c P(X < 76) = 0.2839
P(X < c)
= 0.2839 − 0.2
= 0.0839
c = 70.34 (2 dp)



b P(X < 75) = 0.2375
So P(X < b)
= 0.2375 + 0.4
= 0.6375
b = 82.463



d P(X < Q1) = 0.25
So Q1 = 75.28
P(X < Q3) = 0.75
So Q3 = 84.72
IQR = 84.72 − 75.28
= 9.44

Exercise 3C – Statistics Y2 pages 46-47

Solution 22

6. Badger masses M kg follow $N(4.5, 0.6^2)$.
(a) Find the lower quartile.
(b) Find the 80th percentile.
(c) Explain without calculation why the median is 4.5 kg.

a

$P(M < Q_1) = 0.25$. Inverse normal gives $Q_1 = 4.095$ kg.

b

$P(M < a) = 0.8$ gives $a = 5.005$ kg.

c

The normal distribution is symmetric about its mean. Half the badgers weigh less than 4.5 kg.

Exercise 3C – Statistics Y2 pages 46-47

Solution 23

7. Learner drivers' theory scores X follow $N(72, 6^2)$.

(a) Find a such that $P(X < a) = 0.6$. (1)

(b) Find the interquartile range. (2)

7a

Inverse normal gives $a = 73.52$.

7b

$Q_1 = 67.953\ldots$ and $Q_3 = 76.046\ldots$. Hence $Q_3 - Q_1 = 8.09$.

8. Chocolate bar masses Y g follow $N(60, 2^2)$.

(a) Find y such that $P(Y > y) = 0.2$. (1)

(b) Find the 10% to 90% interpercentile range. (2)

(c) Tom says the median equals the mean. Is he correct? Explain. (1)

8a

$P(Y < y) = 0.8$ gives $y = 61.68$ g.

8b

$62.563\ldots - 57.436\ldots = 5.13$ g.

8c

Yes. A normal distribution is symmetric about its mean.

Exercise 3C – Statistics Y2 pages 46-47

Solution 24

9. Men's heights H cm follow $N(170, 10^2)$. A manufacturer makes frock coats, which reach from neck to near floor.

Short	Regular	Long
30%	50%	20%

(a) Suggest a height range to advertise for regular coats. (4)

(b) State one assumption. (1)

a

The 30th percentile is $170 + 10\Phi^{-1}(0.3) \approx 165$ cm. The 80th percentile is $170 + 10\Phi^{-1}(0.8) \approx 178$ cm. A regular coat suits heights from about 165 cm to 178 cm.

b

Assume customers' heights have the same distribution as the group modelled. Other assumptions concern body proportions and the absence of extreme outliers.

Standard Normal Distribution

Solution 25

Z is the number of standard deviations above the mean.

Using the above definition, and considering the a variable (IQ) distributed as $X \sim N\left(100, 15^2\right)$,calculate (in your head!):

IQ	Z
100	0
130	2
85	-1
165	4.333
62.5	-2.5

What calculation are you actually doing in your head?

For an IQ of 130: $Z = \frac{130-100}{15} = 2$ as expected.

! Z represents the coding:
$Z = \frac{X-\mu}{\sigma}$
and $Z \sim N\left(0, 1^2\right)$. Z is known as a **standard** normal distribution.

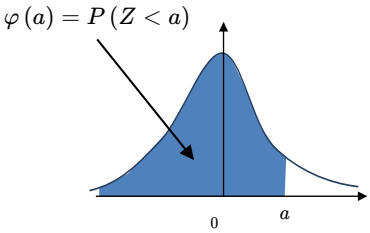
The 0 and 1 of $Z \sim N\left(0, 1^2\right)$ are the result of the coding. If we've subtracted μ from each value the mean of the normal distribution is now 0. If we've divided all the values by σ the standard deviation is now $\frac{\sigma}{\sigma} = 1$

Standard Normal Distribution

Solution 26

The point of coding in this context is that all different possible normal distributions become a single unified distribution where we no longer have to worry about the mean and standard deviation.
It means for example when we calculate $P\left(Z < 3\right)$, this will always give the same probability regardless of the original distribution.

It also means we can look up probabilities in a **z-table**:



$\Phi\left(a\right) = P\left(Z < a\right)$ is the cumulative distribution for the standard normal distribution. The values of $\Phi(a)$ can be found in a z-table.

This is a traditional z-table in the old A Level syllabus (but also found elsewhere). You no longer get given this and are expected to use your calculator.

z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$	z	$\Phi(z)$
0.00	0.5000	0.50	0.6915	1.00	0.8413	1.50	0.9332	2.00	0.9772
0.01	0.5040	0.51	0.6950	1.01	0.8438	1.51	0.9345	2.01	0.9778
0.02	0.5080	0.52	0.6985	1.02	0.8461	1.52	0.9357	2.02	0.9783
0.03	0.5120	0.53	0.7019	1.03	0.8485	1.53	0.9370	2.03	0.9788
0.04	0.5160	0.54	0.7054	1.04	0.8508	1.54	0.9382	2.04	0.9793
0.05	0.5199	0.55	0.7088	1.05	0.8531	1.55	0.9394	2.05	0.9798
0.06	0.5239	0.56	0.7123	1.06	0.8554	1.56	0.9406	2.06	0.9803
0.07	0.5279	0.57	0.7157	1.07	0.8577	1.57	0.9418	2.07	0.9808
0.08	0.5319	0.58	0.7190	1.08	0.8599	1.58	0.9429	2.08	0.9812
0.09	0.5359	0.59	0.7224	1.09	0.8621	1.59	0.9441	2.09	0.9817
0.10	0.5398	0.60	0.7257	1.10	0.8643	1.60	0.9452	2.10	0.9821

This is from the new formula booklet. This is sometimes known as a 'reverse z-table', because you're looking up the z-value for a probability. Beware: p here it the probability of **exceeding** z rather than being up to z. Let's use it...

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

Examples

Solution 27

$X \sim N(50, 4^2)$. Express (a) $P(X < 53)$ and (b) $P(X \geq 55)$ in terms of $\Phi(z)$.

a

$P(X < 53) = P\left(Z < \frac{53-50}{4}\right)$
$= P(Z < 0.75) = \Phi(0.75).$

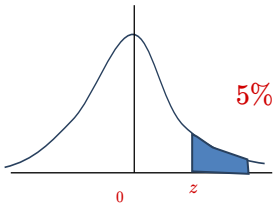
b

$P(X \geq 55) = P\left(Z \geq \frac{55-50}{4}\right)$
$= 1 - P(Z < 1.25) = 1 - \Phi(1.25).$

Standardise to turn X into Z.

Adult systolic blood pressure S mmHg is normal, with mean 127 and standard deviation 16. A researcher studies adults above the 95th percentile. Find the minimum blood pressure for inclusion.

Blood pressure



$S \sim N(127, 16^2)$. The upper 5% cutoff is $z = 1.6449$.
$\frac{s-127}{16} = 1.6449.$
$s = 153$ mmHg (3 s.f.).

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

A sketch helps determine whether z is positive or negative. Once the table gives z, use $z = (s - \mu)/\sigma$.

Further Examples

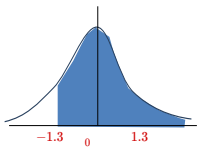
Solution 28

Use the standard normal distribution to answer each question.

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

(a) Find $P(Z > -1.3)$.

a



By symmetry, $P(Z > -1.3) = P(Z < 1.3) = 0.9032$.
Changing both the sign and the inequality preserves the probability.

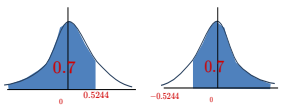
(b) Find $P(-2 < Z < 1)$.

b

$\Phi(1) - \Phi(-2) = 0.8413 - (1 - 0.9772) = 0.8185.$
Subtract the lower cumulative probability from the upper.

(c) Find a if $P(Z > a) = 0.7$.

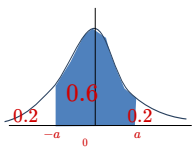
c



The lower tail is 0.3, so a is negative. By symmetry it is the negative of the upper 30% cutoff.
$a = -0.5244.$

(d) Find a if $P(-a < Z < a) = 0.6$.

d



The two tails each have probability 0.2, so $P(Z < a) = 0.8$ and $a = 0.8416$.

Test Your Understanding

Solution 29

1. IQ has mean 100 and standard deviation 15. Use a normal table to find the IQ cutoff for (a) the top 10% and (b) the bottom 20%.

2. $X \sim N(100, 15^2)$. Express in terms of Φ:
(a) $P(X > 115)$
(b) $P(77.5 < X < 112)$.

3. Find a such that:
(a) $P(-a < Z < a) = 0.2$
(b) $P(0 < Z < a) = 0.35$.

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

1a
$z = 1.2816$, so $x = 100 + 15(1.2816) = 119.224$ (about 119.2).

2a
$P(X > 115) = P(Z > 1) = 1 - \Phi(1)$.

3a
Each tail has area 0.4, giving $a = 0.2533$.

1b
$z = -0.8416$, so $x = 100 - 15(0.8416) = 87.376$.

2b
$P(Z < 0.8) - P(Z < -1.5) = \Phi(0.8) + \Phi(1.5) - 1$.

3b
$P(Z > a) = 0.15$, giving $a = 1.0364$.

Exercise 3D – Statistics Y2 pages 49

Solution 30

5. (a) Use a percentage points table to find z such that $P(Z > z) = 0.025$. (1)
(b) A fighter jet training programme takes the top 2.5% of candidates. Test scores are normal with mean 80 and standard deviation 4. Find the necessary score. (2)

a
The upper 2.5% cutoff is $z = 1.96$.

b
$1.96 = (x - 80)/4$ gives $x = 80 + 4(1.96) = 87.84$, or 87.8 (3 s.f.).

Exercise 3D – Statistics Y2 pages 49

Solution 31

6. (a) Use a percentage points table to find z such that $P(Z < z) = 0.15$. (1)

(b) A petite hat should fit the smallest 15% of customers. Hat sizes are normal with mean 57 cm and standard deviation 2 cm. Find the hat size. (2)

a

The upper-tail table gives 1.0364. By symmetry the lower-tail cutoff is -1.0364.

b

$x = 57 + 2(-1.0364) = 54.9272$, or 54.9 cm (3 s.f.).

Exercise 3D – Statistics Y2 pages 49

Solution 32

7. (a) Use a percentage points table to find the z-values for the 10% to 90% interpercentile range. (2)

(b) Bulb life is normal with mean 1175 hours and standard deviation 56 hours. A standard bulb's life lies in that range. Find the endpoints to the nearest hour. (2)

a

By symmetry the cutoffs are -1.2816 and 1.2816.

b

$1175 \pm 56(1.2816) = 1175 \pm 71.7696$. The life range is 1103 to 1247 hours.

Finding the missing mean or standard deviation

Solution 33

In the last section, you may have thought, "what's the point of standardising to Z when I can just use the DISTRIBUTION mode on my calculator?"

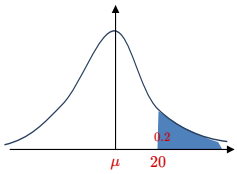
HINT: both forward and reverse normal lookups on the calculator **required you to specify μ and σ.**

[Textbook] $X \sim N\left(\mu, 3^2\right)$. Given that $P\left(X > 20\right) = 0.2$, find the value of μ.

$P\left(X > 20\right) = 0.2$

$0.8416 = \frac{20-\mu}{3}$

$\mu = 17.5 (3sf)$

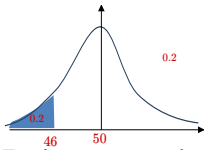


In the last section, you may have thought, "what's the point of standardising to Z when I can just use the DISTRIBUTION mode on my calculator?"

[Textbook] A machine makes metal sheets with width, X cm, modelled as a normal distribution such that $X \sim N\left(50, \sigma^2\right)$.

(a) Given that $P\left(X < 46\right) = 0.2119$, find the value of σ.

(b) Find the 90^{th} percentile of the widths.



a $P(X < 46) = 0.2119$
$-0.80 = (46 - 50)/\sigma$
$\sigma = 5$

b $X \sim N(50, 5^2)$
$P(X < a) = 0.9$
$a = 56.4$ cm (1 dp)

For inverse normal on the calculator, $Z \sim N(0, 1^2)$: use $\mu = 0$ and $\sigma = 1$.

The method here is exactly the same as before:

1. Using a sketch, determine whether you're expecting a positive or negative z value.
2. Look up z value, using tables if you can (otherwise your calculator). Make negative if in bottom half.
3. Use $Z = \frac{X-\mu}{\sigma}$

When both are missing

Solution 34

Edexcel S1 Jan 2011. Machine B fills cartons with soup. The mass Y grams is normal with mean μ and standard deviation σ.

(c) Given $P(Y < 160) = 0.99$ and $P(Y > 152) = 0.90$, find μ and σ. (6)

(c)

$160 - \mu = 2.3263\sigma$ and $152 - \mu = -1.2816\sigma$

.

Subtract: $8 = 3.6079\sigma$. Hence $\sigma \approx 2.22$ g and $\mu \approx 154.84$ g.

Test Your Understanding

Solution 35

Edexcel S1 May 2013(R). London–Berlin flight times are normal with mean 100 minutes and standard deviation d.

(b) 15% take more than 115 minutes. Find d. (4)

b

$P(B > 115) = 0.15$ gives $(115 - 100)/d = 1.0364$. Thus $d \approx 14.5$ minutes.

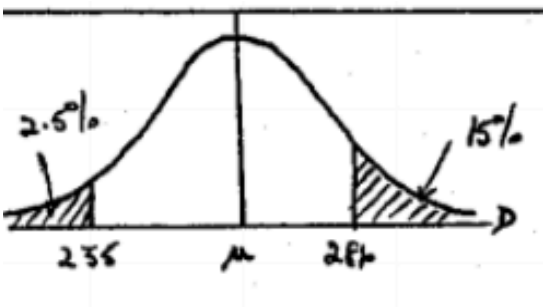
Edexcel S1 Jan 2002 Q5. Pregnancy durations for a breed of cow are normal. 2.5% are shorter than 235 days and 15% exceed 286 days.

(a) Show $\mu - 235 = 1.96\sigma$. (2)

(b) Obtain a second equation. (3)

(c) Find μ, σ. (4)

(d) Find the limits containing the middle 68.3%. (2)



5a $(235 - \mu)/\sigma = -1.96$, hence $\mu - 235 = 1.96\sigma$.

5b

$(286 - \mu)/\sigma = 1.0364$.

5c

$51 = (1.96 + 1.0364)\sigma$, so $\sigma \approx 17.02$, $\mu \approx 268.36$ days.

5d

$\mu \pm \sigma$ gives about 251 to 285 days.

Exercise 3E – Statistics Y2 pages 51-53

Solution 36

11. Pottery bowl diameters D mm are normal with standard deviation 5 mm. 75% exceed 200 mm.

(a) Find the mean μ. (2)

(b) Find $P(204 < D < 206)$. (1)

(c) Three bowls are selected at random. Find the probability all exceed 205 mm. (3)

a

$P(D < 200) = 0.25$, so $(200 - \mu)/5 = -0.67449\ldots$. Hence $\mu = 203.372\ldots$ mm.

b

$\Phi(0.5255\ldots) - \Phi(0.1255\ldots) \approx 0.1504$.

c

$P(D > 205) = 1 - \Phi(0.3255\ldots) \approx 0.3724$.

By independence the required probability is $0.3724\ldots^3 \approx 0.0516$.

Exercise 3E – Statistics Y2 pages 51-53

Solution 37

12. Cloth thickness T mm is normal with mean 2.5 mm. 65% of cloths are thinner than 2.55 mm.
(a) Find the standard deviation. (2)
(b) Find the proportion between 2.4 mm and 2.6 mm thick. (1)
(c) Only cloths in this range can be sold. Find the probability at least 15 of a random sample of 20 can be sold. (3)

a

$(2.55 - 2.5)/\sigma = \Phi^{-1}(0.65) = 0.38532\ldots$.
Hence $\sigma = 0.1298$ mm.

b

$\Phi(0.77065\ldots) - \Phi(-0.77065\ldots) = 0.5591$.

c

With $Y \sim B(20, 0.5591\ldots)$, $P(Y \geq 15) = 1 - P(Y \leq 14) = 0.0650$.

Exercise 3E – Statistics Y2 pages 51-53

Solution 38

13. Penguin masses M kg are normal with mean μ and standard deviation σ. 10% weigh under 18 kg and 5% over 30 kg.
(a) Sketch this information. (2)
(b) Find μ, σ. (6)
(c) Ten penguins are chosen at random. Find the probability at least four exceed 25 kg. (4)

a Shade 0.10 below 18 and 0.05 above 30.

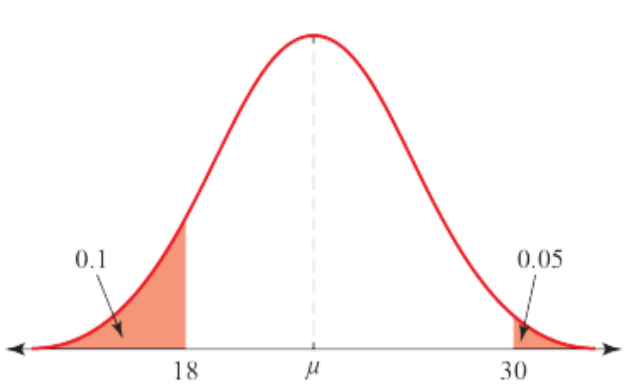
b

$18 - \mu = -1.28155\sigma$ and $30 - \mu = 1.64485\sigma$. Subtraction gives $12 = 2.92640\sigma$, so $\sigma = 4.101$, $\mu = 23.26$ (4 s.f.).

c

$p = P(M > 25) = 1 - \Phi((25 - 23.2551)/4.1006) \approx 0.33522$.
$Y \sim B(10, p)$ gives $P(Y \geq 4) = 1 - P(Y \leq 3) = 0.4459$.

a



Exercise 3E – Statistics Y2 pages 51-53

Solution 39

14. Adult dachshund lengths are normal. 20% are shorter than 16 inches and 10% exceed 18 inches.

(a) Find μ and σ. (6)

(b) Find the interquartile range. (2)

a

$16 - \mu = -0.84162\sigma$, $18 - \mu = 1.28155\sigma$. Subtract: $2 = 2.12317\sigma$. Thus $\sigma = 0.9420$ and $\mu = 16.79$ inches (4 s.f.).

b

The quartiles use $z = \pm 0.67449$. $Q_1 = 16.1574\ldots$, $Q_3 = 17.4281\ldots$. The IQR is 1.27 inches.

EXTENSION – Challenge P53

Solution 40

$X \sim N(\mu, \sigma^2)$ has interquartile range q.

(a) Express σ as a multiple of q, to 3 s.f.

(b) Explain why μ cannot be expressed in terms of q alone.

a

$z = \Phi^{-1}(0.75) = 0.674489\ldots$. The quartiles are $\mu \pm z\sigma$, hence $q = 2z\sigma$. $\sigma = q/(2z) = 0.741q$ (3 s.f.).

b

The mean cancels when subtracting the quartiles, so q contains no information about μ.

Knowledge Recall on Binomial Distribution

Solution 41

$X \sim B(n, p)$ counts successes in n trials, each with success probability p. What conditions are required?

The number of trials is fixed.

Each trial has two possible outcomes, success or failure.

The probability of success is constant.

The trials are independent.

$P(X = r) = \binom{n}{r} p^r (1-p)^{n-r}$.

Knowledge Recall on Binomial Distribution

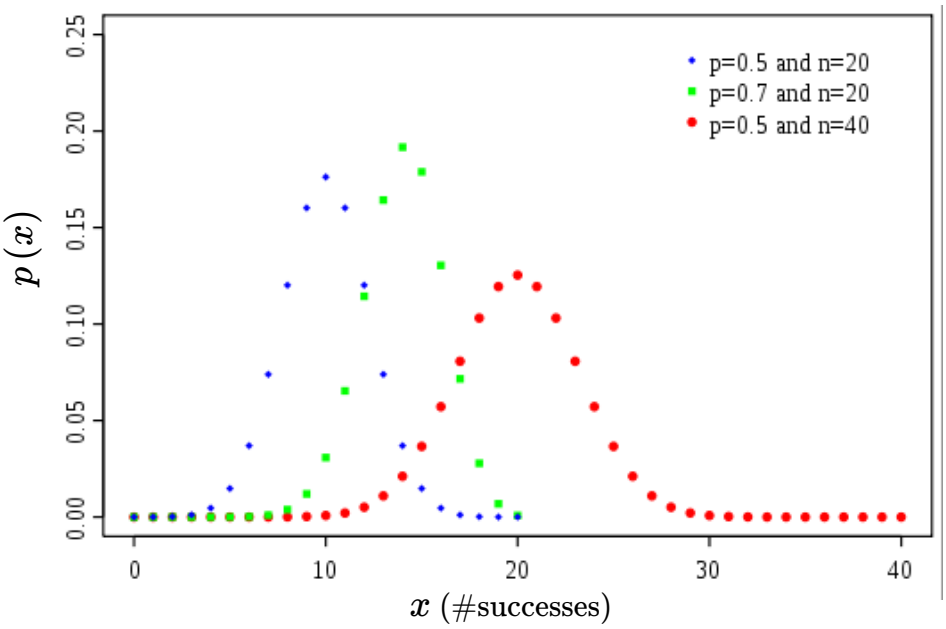
Solution 42

Examples of binomial counts

1. Flawed garments among 50 sampled.
2. Votes in favour of a new initiative among 3000 sampled voters, assuming independent decisions.
3. Left-handed students in a class of 34.

Approximating a Binomial Distribution

Solution 43



The graph shows several binomial probability functions. Which one resembles another distribution, and which distribution does it resemble?

With p close to 0.5 and n fairly large, the shape resembles a normal distribution. At $p = 0.5$ the binomial distribution is symmetric: one head in ten tosses is as likely as one tail.

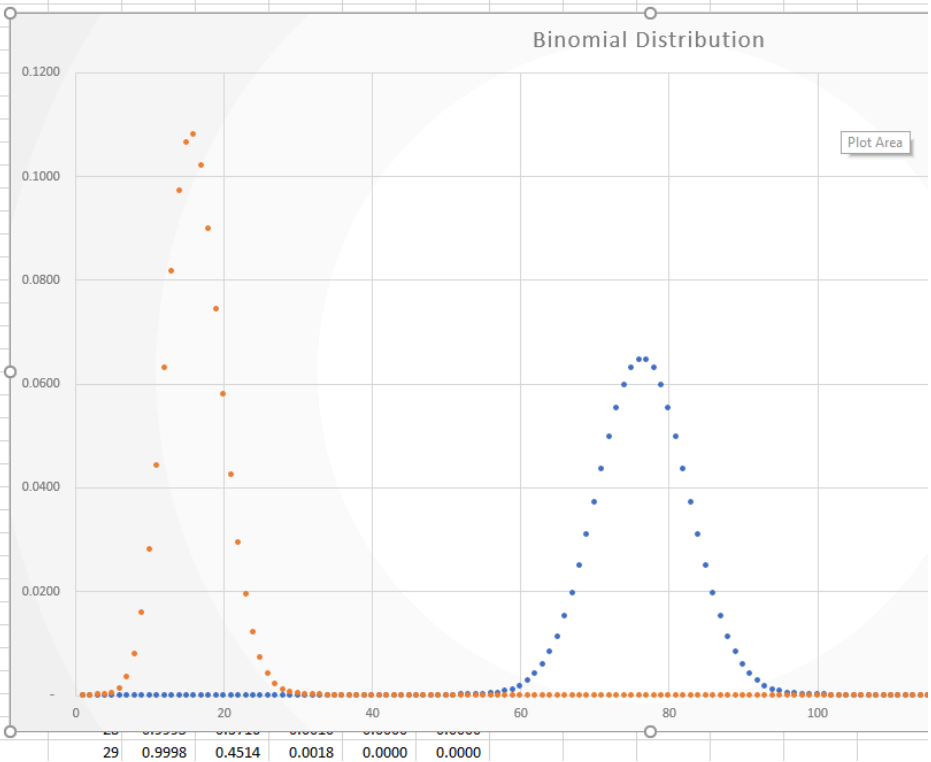
Modelling binomial distribution in excel

Solution 44

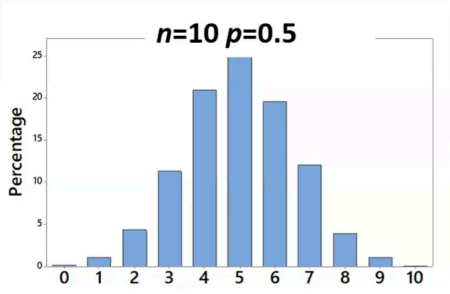
Background curiosity.. How to plot a binomial distribution in Excel

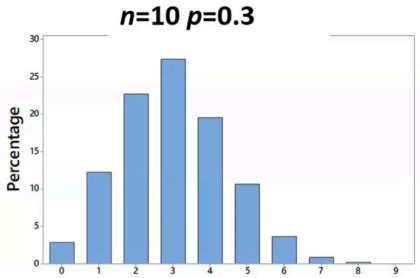
Microsoft Excel Worksheet

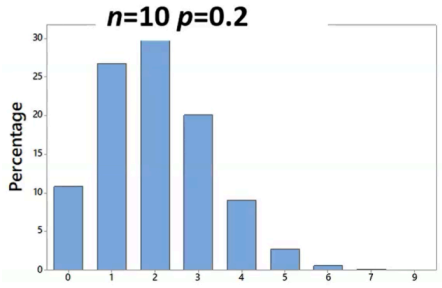
	Normal				
r	0.1	0.2	0.3	0.4	0.5
0	0.0000	0.0000	0.0000	0.0000	0.0000
1	0.0000	0.0000	0.0000	0.0000	0.0000
2	0.0000	0.0000	0.0000	0.0000	0.0000
3	0.0001	0.0000	0.0000	0.0000	0.0000
4	0.0004	0.0000	0.0000	0.0000	0.0000
5	0.0013	0.0000	0.0000	0.0000	0.0000
6	0.0035	0.0000	0.0000	0.0000	0.0000
7	0.0079	0.0000	0.0000	0.0000	0.0000
8	0.0159	0.0000	0.0000	0.0000	0.0000
9	0.0280	0.0000	0.0000	0.0000	0.0000
10	0.0443	0.0000	0.0000	0.0000	0.0000
11	0.0630	0.0000	0.0000	0.0000	0.0000
12	0.0817	0.0000	0.0000	0.0000	0.0000
13	0.0971	0.0001	0.0000	0.0000	0.0000
14	0.1063	0.0002	0.0000	0.0000	0.0000
15	0.1079	0.0004	0.0000	0.0000	0.0000
16	0.1019	0.0008	0.0000	0.0000	0.0000
17	0.0899	0.0016	0.0000	0.0000	0.0000
18	0.0744	0.0031	0.0000	0.0000	0.0000
19	0.0578	0.0054	0.0000	0.0000	0.0000
20	0.0424	0.0089	0.0000	0.0000	0.0000
21	0.0294	0.0138	0.0000	0.0000	0.0000
22	0.0193	0.0204	0.0000	0.0000	0.0000
23	0.0120	0.0286	0.0000	0.0000	0.0000
24	0.0071	0.0381	0.0000	0.0000	0.0000
25	0.0040	0.0484	0.0001	0.0000	0.0000
26	0.0022	0.0587	0.0001	0.0000	0.0000
27	0.0011	0.0679	0.0002	0.0000	0.0000
28	0.0005	0.0752	0.0005	0.0000	0.0000
29	0.0003	0.0797	0.0009	0.0000	0.0000
29	0.9998	0.4514	0.0018	0.0000	0.0000



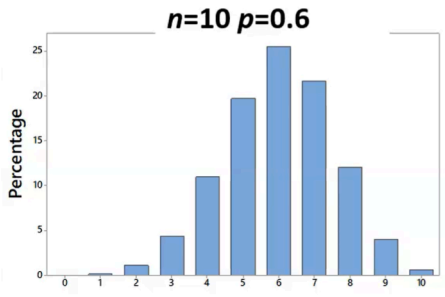
As p decreases the distribution becomes more positively skewed

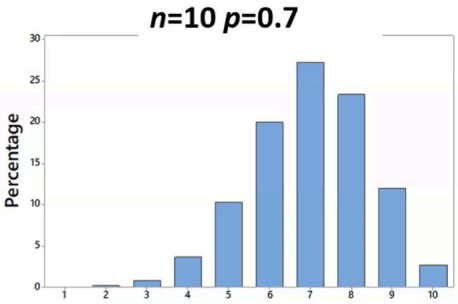


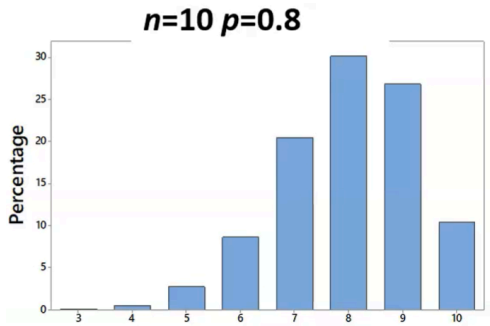




As p increases the distribution becomes more negatively skewed







If we're going to use a normal distribution to approximate a Binomial distribution, it makes sense that we set the mean and standard deviation of the normal distribution to match that of the original binomial distribution:

$\mu =$ np

$\sigma =$ $\sqrt{np(1-p)}$

If n is large and p close to 0.5, then the binomial distribution $X \sim B(n, p)$ can be approximated by the normal distribution $N(\mu, \sigma^2)$ where

$\mu = np$

$\sigma = \sqrt{np(1-p)}$

Quickfire Questions:

$X \sim B(10, 0.2) \rightarrow Y \sim$ $N(2, 1.6)$

$X \sim B(20, 0.5) \rightarrow Y \sim$ $N(10, 5)$

$X \sim B(6, 0.3) \rightarrow Y \sim$ $N(1.8, 1.26)$

We tend to use the letter Y to represent the normal distribution approximation of the distribution X.

Why use a normal approximation?

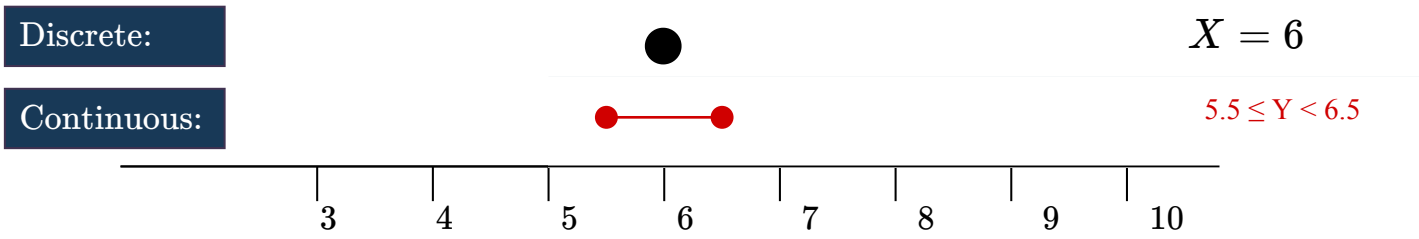
- Tables for the binomial distribution only goes up to $n = 50$ and your calculator will reject large values of n.
- The formula for $P(X = x)$ makes use a factorials. Factorials of large numbers cannot be computed efficiently. Type in 65! for example; your calculator will hesitate! Now imagine how many factorials would be required if you wanted to find $P(X \leq 65)$.

Continuity Corrections

Solution 47

One problem is that the outcomes of a binomial distribution (i.e. number of successes) are **discrete** whereas the Normal distribution is **continuous**.
We apply something called a **continuity correction** to approximate a discrete distribution using a continuous one.

The random variable X represents the time to finish a race in hours. We're interested in knowing the probability Alice took 6 hours to the nearest hour.
How would you represent this time on a number line given hours is discrete? And what about if hours was now considered to be continuous (as Y)?

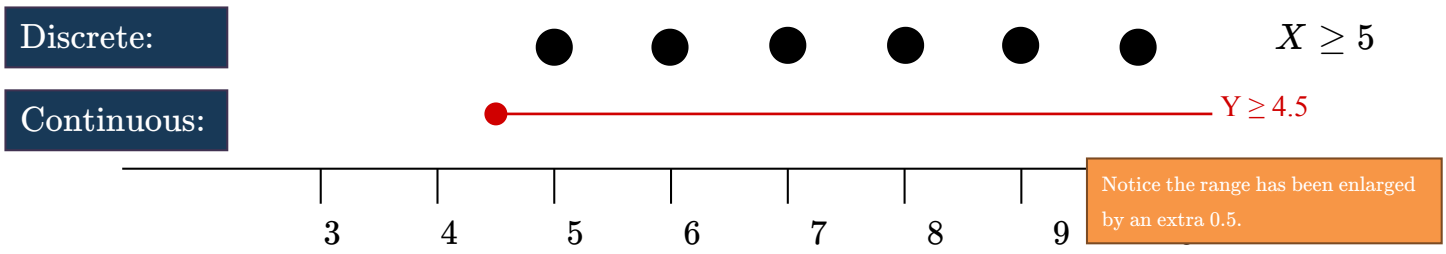


We can't just find $\mathbf{P}(\mathbf{Y} = \mathbf{6})$ when $\mathbf{Y}$ is continuous, because the probability is effectively 0. But $\mathbf{P}(\mathbf{5.5} < \mathbf{Y} < \mathbf{6.5})$ would seem a sensible interval to use because any time between 5.5 and 6.5 would have rounded to 6 hours were it discrete.

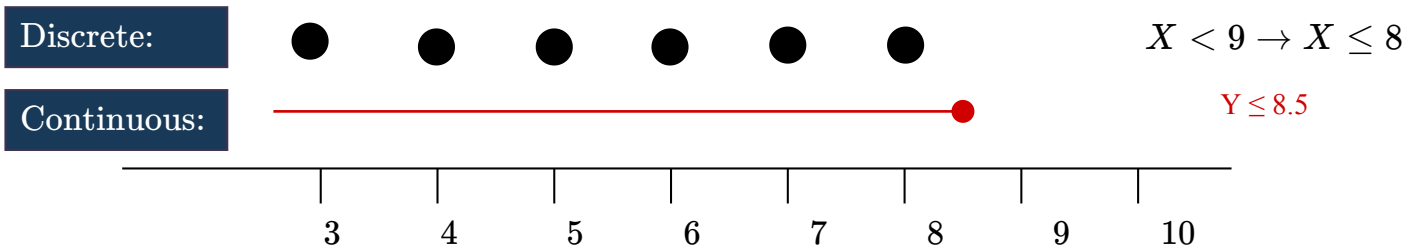
Continuity Corrections

Solution 48

If X is a discrete variable, and Y is its continuous equivalent, how would you represent $P(X \geq 5)$ for Y?



How would represent $P(X < 9)$ for Y?



A continuity correction is approximating a discrete range using a continuous one.
1. If $>$ or $<$, convert to $\geq$, $\leq$ first.
2. Enlarge the range by 0.5.

Examples

Solution 49

Discrete probability	Continuous approximation
$P(X \leq 7)$	$\approx P(Y \leq 7.5)$
$P(X < 10)$	$= P(X \leq 9) \approx P(Y \leq 9.5)$
$P(X > 9)$	$= P(X \geq 10) \approx P(Y \geq 9.5)$
$P(1 \leq X \leq 10)$	$\approx P(0.5 \leq Y \leq 10.5)$
$P(3 < X < 6)$	$= P(4 \leq X \leq 5) \approx P(3.5 \leq Y \leq 5.5)$
$P(3 \leq X < 6)$	$\approx P(2.5 \leq Y \leq 5.5)$
$P(3 < X \leq 6)$	$\approx P(3.5 \leq Y \leq 6.5)$
$P(X = 3)$	$\approx P(2.5 \leq Y \leq 3.5)$

Continuity correction. First rewrite strict inequalities using integer endpoints and $\leq$ or $\geq$. Then extend the discrete range by 0.5 at each end.

Comparison on your calculators

Solution 50

400 children guess true or false on each of 80 quantum physics questions. What proportion of children get fewer than 38 correct?

1. Find the normal distribution

Let X be the number correct. $X \sim B(80, 0.5)$.

The normal approximation is $Y \sim N(40, 20)$.

2. Adjust for continuity

$P(X < 38) = P(X \leq 37) \approx P(Y < 37.5)$.

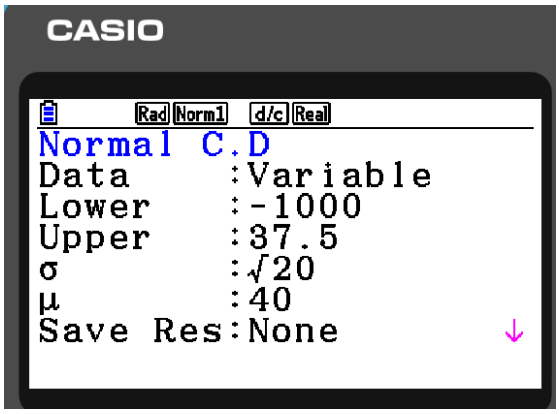
Comparison on your calculators — Solution 51

400 children guess true or false on each of 80 quantum physics questions. What proportion of children get fewer than 38 correct?

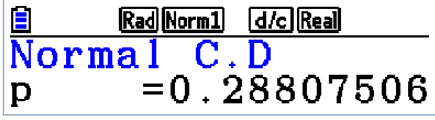
Normal Distribution

Write distribution

Y ~ N(40, 20); find P(Y < 37.5).



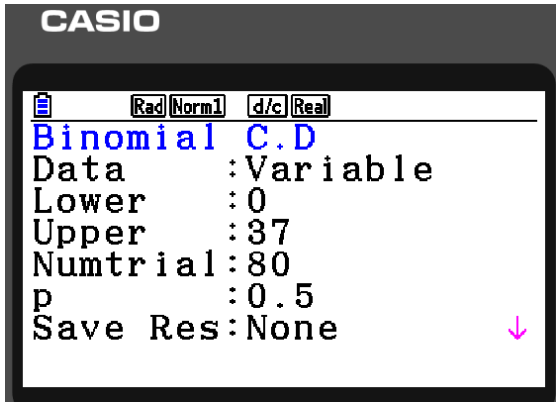
Display result



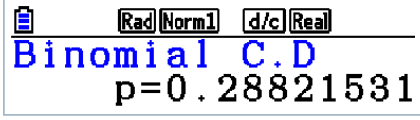
Binomial Distribution

Write distribution

X ~ B(80, 0.5); find P(X ≤ 37).



Display result



Full Example — Solution 52

[Textbook - Edited] For a particular type of flower bulbs, 55% will produce yellow flowers. A random sample of 80 bulbs is planted.

(a) Calculate the actual probability that there are exactly 50 flowers.

(b) Use a normal approximation to find a estimate that there are exactly 50 flowers.

(c) Hence determine the percentage error of the normal approximation for 50 flowers.

a

a $X \sim B(80, 0.55)$

$P(X = 50) = C(80, 50) \times 0.55^{50} \times 0.45^{30} = 0.0365$

b

b $Y \sim N(44, 19.8)$

$P(X = 50) \approx P(49.5 < Y < 50.5)$

$= 0.9280 - 0.8918 = 0.0362$ (4 dp)

c

c

Percentage error:

$$\frac{0.0365 - 0.0362}{0.0365} \times 100 = 0.82\%$$

Test Your Understanding

Solution 53

Edexcel S2 Jan 2004 Q3. $X \sim B(n, p)$.

(a) Which p gives the most accurate normal approximation? (1)

(b) Give a reason. (1)

(c) With $n = 200, p = 0.48$, find $P(90 \leq X < 105)$. (7)

a

$p = \frac{1}{2}$.

b

The binomial distribution is then symmetric.

c

$Y \sim N(96, 49.92)$ since $np = 96$ and $np(1 - p) = 49.92$.

Use $P(89.5 < Y < 104.5)$. The standardised limits are approximately -0.92 and 1.20. The probability is about 0.706.

Exercise 3F – Statistics Y2 pages 55

Solution 54

2. $X \sim B(150, 0.45)$. Use a suitable approximation to estimate:

(a) $P(X \leq 60)$ (b) $P(X > 75)$ (c) $P(65 \leq X \leq 80)$

2a 0.1253

2b 0.0946

2c 0.6723

3. $X \sim B(200, 0.53)$. Use a suitable approximation to estimate:

(a) $P(X < 90)$ (b) $P(100 \leq X < 110)$ (c) $P(X = 105)$

3a 0.0097

3b 0.5115

3c 0.0559

4. $X \sim B(100, 0.6)$. Use a suitable approximation to estimate:

(a) $P(X > 58)$ (b) $P(60 < X \leq 72)$ (c) $P(X = 70)$

4a 0.6203

4b 0.4540

4c 0.0102

5. A fair coin is tossed 70 times. Estimate the probability of more than 45 heads using a suitable approximation.

5
0.006

6. A roulette ball lands on red with probability $\frac{50}{101}$. Estimate the probability of red in at least half of 1200 spins.

6
0.3767

Exercise 3F – Statistics Y2 pages 55

Solution 55

7. (a) Give two conditions for a normal approximation to the binomial. (2)

45% of orchids have pink flowers. In a random sample of 20, X are pink.

(b) Find $P(X = 10)$. (1)

A second random sample contains 240 orchids.

(c) Estimate the probability fewer than 110 are pink. (3)

(d) The probability at least q are pink is 0.2. Find q. (3)

a

The number of trials should be large and the probability not too close to 0 or 1. Here 240 is large and 0.45 is close to one half.

b

$\binom{20}{10}(0.45)^{10}(0.55)^{10} = 0.1593.$

c

$Y \sim N(108, 59.4)$ since $np = 108$ and $np(1-p) = 59.4$.

$P(X < 110) \approx P(Y < 109.5) = 0.5772.$

d

$P(Y > q - 0.5) = 0.2$. Inverse normal gives $q - 0.5 = 114.485$, so $q = 115$.

Exercise 3F – Statistics Y2 pages 55

Solution 56

8. 52% of drill bits last over 40 hours. Of 30 random bits X last over 40 hours.

(a) Find $P(X < 17)$. (1)

(b) Of 600 random bits, estimate the probability between 300 and 350 inclusive last over 40 hours. (3)

8a

$X \sim B(30, 0.52)$; $P(X \leq 16) = 0.6277.$

8b

$Y \sim N(312, 149.76)$. By continuity correction $P(299.5 < Y < 350.5) = 0.8456.$

9. 56% of cereal boxes contain prizes. In a random sample of 100 boxes:

(a) Find the exact probability that 55 contain prizes. (1)

(b) Find the percentage error when a normal approximation is used. (4)

9a

$\binom{100}{55}(0.56)^{55}(0.44)^{45} = 0.07838\ldots.$

9b

$Y \sim N(56, 24.64)$. $P(54.5 < Y < 55.5) = 0.07863\ldots$. The approximation overestimates by about 0.31%.

Distribution of Sample means

Solution 57

Choose three numbers independently and uniformly from the integers 1 to 100. One sample is 55, 74, 5. What is its mean?

Sample mean

$\bar{x} = (55 + 74 + 5)/3 = 44.\overline{6}.$

What are the smallest and largest possible sample means?

Smallest

1, for the sample $\{1, 1, 1\}$.

Largest

100, for the sample $\{100, 100, 100\}$.

Which sample means are most likely? Which are least likely? What shape would the distribution have if we calculated the probability of every possible sample mean?

Roll of a fair die: sample size 1

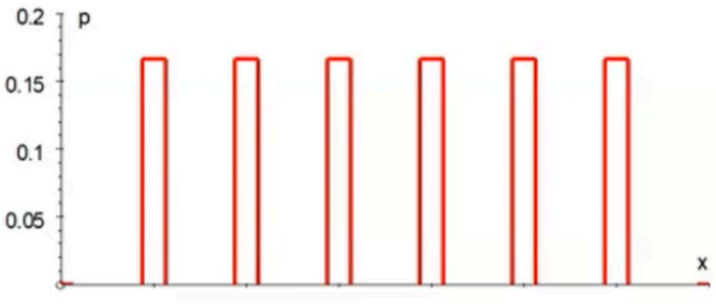
Solution 58

Roll one fair die. The population is $\{1, 2, 3, 4, 5, 6\}$. List the possible samples of size 1.

Samples $\{1\}, \{2\}, \{3\}, \{4\}, \{5\}, \{6\}$.

Probability distribution of the sample means

$\bar{x}$	1	2	3	4	5	6
$P(\bar{X} = \bar{x})$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$



Expectation

$E(\bar{X}) = 3.5.$

Variance

$\mathrm{Var}(\bar{X}) = \frac{35}{12}.$

With a sample size of 1, the mean and variance equal those of the population.

Roll of a fair die: sample size 2 — Solution 59

Roll a fair die twice. How many ordered samples of size two are possible? What are their means?

Sample means

There are $6^2 = 36$ equally likely ordered pairs.

	1	2	3	4	5	6
1	1	1.5	2	2.5	3	3.5
2	1.5	2	2.5	3	3.5	4
3	2	2.5	3	3.5	4	4.5
4	2.5	3	3.5	4	4.5	5
5	3	3.5	4	4.5	5	5.5
6	3.5	4	4.5	5	5.5	6

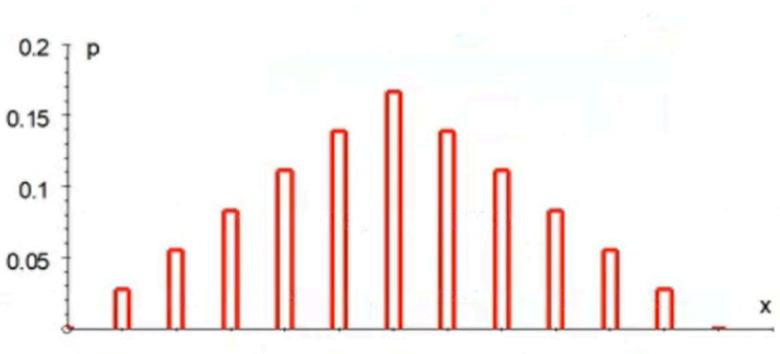
Distribution

$\bar{y}$	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6
$P(\bar{Y} = \bar{y})$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Roll of a fair die: sample size 2 — Solution 60

Distribution of the sample mean for two rolls of a fair die

$\bar{y}$	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6
$P(\bar{Y} = \bar{y})$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$



Expectation

$E(\bar{Y}) = 3.5.$

Variance

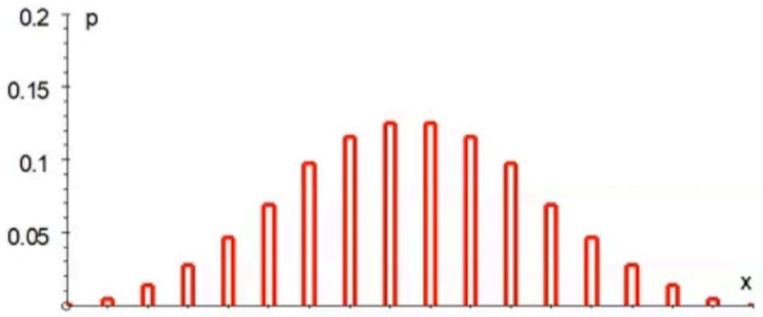
$\mathrm{Var}(\bar{Y}) = 35/24.$

Roll of a fair die: sample size 3

Solution 61

Roll a fair die three times. Find the distribution of the sample mean, its expectation and its variance.

$\bar{z}$	1	$1\frac{1}{3}$	$1\frac{2}{3}$	2	$2\frac{1}{3}$	$2\frac{2}{3}$	3	$3\frac{1}{3}$	$3\frac{2}{3}$	4	$4\frac{1}{3}$	$4\frac{2}{3}$	5	$5\frac{1}{3}$	$5\frac{2}{3}$	6
$P(\bar{Z}=\bar{z})$	$\frac{1}{216}$	$\frac{3}{216}$	$\frac{6}{216}$	$\frac{10}{216}$	$\frac{15}{216}$	$\frac{21}{216}$	$\frac{25}{216}$	$\frac{27}{216}$	$\frac{27}{216}$	$\frac{25}{216}$	$\frac{21}{216}$	$\frac{15}{216}$	$\frac{10}{216}$	$\frac{6}{216}$	$\frac{3}{216}$	$\frac{1}{216}$



Expectation

$E(\bar{Z}) = 3.5.$

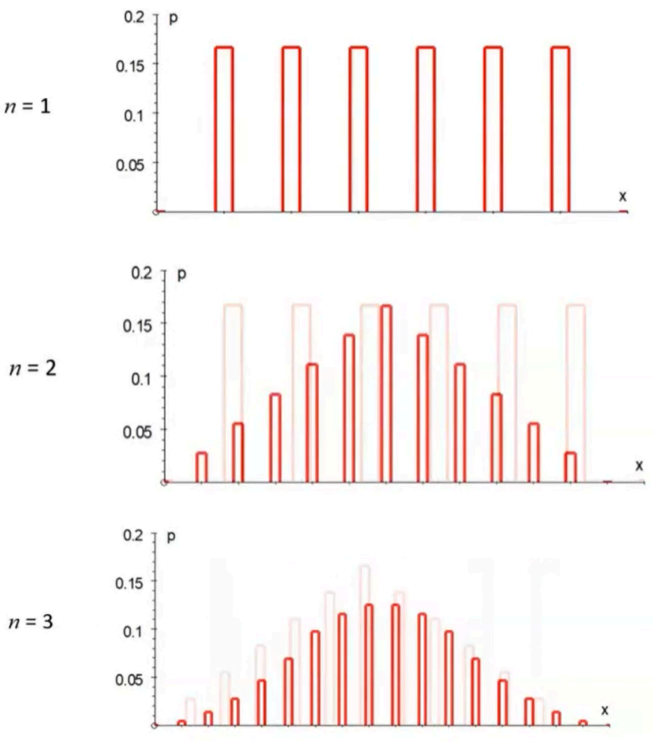
Variance

$\mathrm{Var}(\bar{Z}) = \frac{35}{36}.$

Pattern relating to the sample size

Solution 62

What changes as the sample size increases?



As the sample size increases, the distribution of the sample mean becomes more bell-shaped.

How do mean and spread depend on sample size?

n	Mean	Variance	Standard deviation
1	3.5	$\frac{35}{12}$	$\sqrt{\frac{35}{12}}$
2	3.5	$\frac{35}{24}$	$\sqrt{\frac{35}{24}}$
3	3.5	$\frac{35}{36}$	$\sqrt{\frac{35}{36}}$

Pattern relating to the sample size

Solution 63

How do mean, variance and standard deviation change as sample size increases?

n	Mean	Variance	Standard deviation
1	3.5	$\frac{35}{12}$	$\sqrt{\frac{35}{12}}$
2	3.5	$\frac{35}{24}$	$\sqrt{\frac{35}{24}}$
3	3.5	$\frac{35}{36}$	$\sqrt{\frac{35}{36}}$

← **Sample size 1**
Mean $= \mu$, SD $= \sigma$

← **Sample size 2**
Mean $= \mu$, SD $= \sigma/\sqrt{2}$

← **Sample size 3**
Mean $= \mu$, SD $= \sigma/\sqrt{3}$

Summary

Solution 64

For independent observations with population mean μ and standard deviation σ:

1. The mean of the sample means is μ.
2. The standard deviation of sample means (the standard error) is $\sigma/\sqrt{n}$. It decreases as n increases.

For a random sample of size n from a normally distributed population X, the sample mean $\overline{X}$ is normally distributed with $\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

Hypothesis Testing on the Sample Mean

Solution 65

If the population is $N(\mu, \sigma^2)$, the mean of an independent random sample of size n also follows a normal distribution.

Individual observation

$X \sim N(\mu, \sigma^2)$

$z = \frac{x-\mu}{\sigma}$.

Sample mean

$\bar{X} \sim N(\mu, \sigma^2/n)$.

$z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$.

Calculator skills (Casio fx-CG50): Normal hypothesis testing

Set up of Hypothesis test

Solution 66

4 steps for setting up a Hypothesis Test for the population mean:

1. State $H_0 : \mu = m$, where m is the proposed population mean.

2. Choose the alternative: $\mu > m$ or $\mu < m$ for a one-tailed test, or $\mu \neq m$ for a two-tailed test.

3. Determine the significance level.

4. Consider the evidence from the sample.

Method 1 using a p-value

Solution 67

Test results have mean 65 and standard deviation 10 and are normally distributed. After a new teacher arrives, eight students have mean score 72. Is there evidence of improvement at the 5% significance level?

$X \sim N(65, 10^2)$ under the null hypothesis.

$H_0 : \mu = 65$, where μ is the population mean score.

$H_1 : \mu > 65$.

$\bar{X} \sim N(65, 10^2/8)$.

The distribution of the sample mean is $\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

$P(\bar{X} > 72) = P\left(Z > \frac{72-65}{10/\sqrt{8}}\right)$.

$$Z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$P(Z > 1.98) = 1 - P(Z < 1.98)$

$= 1 - 0.9761 = 0.0239$.

Method 1 using a p-value

Solution 68

Test scores are normal with mean 65 and standard deviation 10. After a new teacher arrives, eight students have a mean of 72. Test at 5% whether the population mean has increased.

In summary:

$H_0 : \mu = 65$, $H_1 : \mu > 65$, where μ is the population mean score.
Under H_0, $\bar{X} \sim N(65, 10^2/8)$ and $P(\bar{X} > 72) = 0.0239$.
$0.0239 < 0.05$, so reject H_0. There is evidence that the mean score has increased.

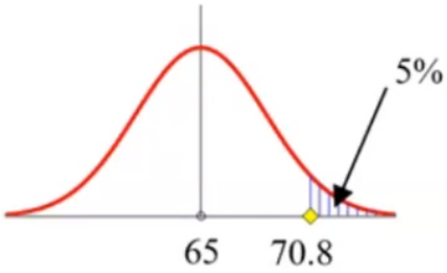
Method 2 Finding a critical region using a p-value

Solution 69

Test scores are normal with mean 65 and standard deviation 10. After a new teacher arrives, eight students have a mean of 72. Test at 5% whether the population mean has increased.

$$\bar{X} \sim N(65, 10^2/8)$$

Using your calculator you find that the critical region is $\bar{x} > 70.8$



The observed sample mean $\bar{x} = 72$ lies in the critical region. Reject H_0: there is evidence that the mean score has increased.

Example 1: Left Hand Tail using a p-Value

Solution 70

The supplier of LITE light bulbs claims a mean life of 130 hours. A trading standards organisation tests 400 bulbs after customer complaints. Their mean lifetime is 128.5 hours and the sample standard deviation is 13 hours. Test at the 2% level whether the population mean lifetime is lower than 130 hours.

Let X be the lifetime of a LITE bulb.

Mean Under H_0, the sampling distribution is centred at 130.

Variance $\bar{X} \approx N(130, 13^2/400)$. The sample standard deviation estimates the population standard deviation.

Hypotheses $H_0 : \mu = 130, H_1 : \mu < 130$.

p-value
$P(\bar{X} < 128.5) \approx 0.0105$.

Conclusion
$0.0105 < 0.02$. Reject H_0: there is evidence that the mean lifetime is below 130 hours.

Example 2: Left Hand Tail using a critical region

Solution 71

The supplier of LITE light bulbs claims a mean life of 130 hours. A trading standards organisation tests 400 bulbs after customer complaints. Their mean lifetime is 128.5 hours and the sample standard deviation is 13 hours. Test at the 2% level whether the population mean lifetime is lower than 130 hours.

1. Let X be a bulb's lifetime. $H_0 : \mu = 130$, $H_1 : \mu < 130$.
Under H_0, $\bar{X} \approx N(130, 13^2/400)$.

2. Use inverse normal for the lower 2% tail.

Critical value

The lower 2% point is 128.67 hours.

Critical region

$\bar{X} < 128.67$.

Conclusion

128.5 lies in the critical region. Reject the null hypothesis. The sample gives evidence that mean lifetime is below 130 hours.

Left Hand Tail: Method 2 using a critical region

Solution 72

The supplier of LITE light bulbs claims a mean life of 130 hours. A trading standards organisation tests 400 bulbs after customer complaints. Their mean lifetime is 128.5 hours and the sample standard deviation is 13 hours. Test at the 2% level whether the population mean lifetime is lower than 130 hours.

1. Let X be a bulb's lifetime. $H_0 : \mu = 130$, $H_1 : \mu < 130$.
Under H_0, $\bar{X} \approx N(130, 13^2/400)$.

2. Use inverse normal for the lower 2% tail.

Critical value

The lower 2% point is 128.67 hours.

Critical region

$\bar{X} < 128.67$.

Conclusion

128.5 lies in the critical region. Reject the null hypothesis. The sample gives evidence that mean lifetime is below 130 hours.

Two tailed tests

Solution 73

In a one-tailed test the alternative specifies a direction: $\mu < k$ or $\mu > k$.

In a two-tailed test $H_1 : \mu \neq k$. A difference in either direction is relevant.

There are two parts to the critical region (both tail of the distribution are considered)

- At the 5% significant level the critical region will have 2.5% in each tail
- When considering the p values compare the value with half of the significance level as you are just comparing with one of the tails

Two tailed test example

Solution 74

Leaves of a rare plant species have normal lengths with mean 8.6 cm and standard deviation 1.2 cm. A botanist finds a clump of plants and measures 50 leaves. Their lengths total 442 cm. Test at the 5% level whether the mean differs from 8.6 cm. What should the botanist conclude?

Let X be leaf length. Under H_0, $\bar{X} \sim N(8.6, 1.2^2/50)$. In general the sample mean has variance σ^2/n.

$H_0 : \mu = 8.6, H_1 : \mu \neq 8.6.$

Use the sample mean of the 50 measured leaves.

$\bar{x} = 442/50 = 8.84$ cm.

$P(\bar{X} > 8.84) = 1 - 0.92135\ldots = 0.0787.$

$0.0787 > 0.025$, so do not reject H_0. The results do not establish that the plants are of a different species.

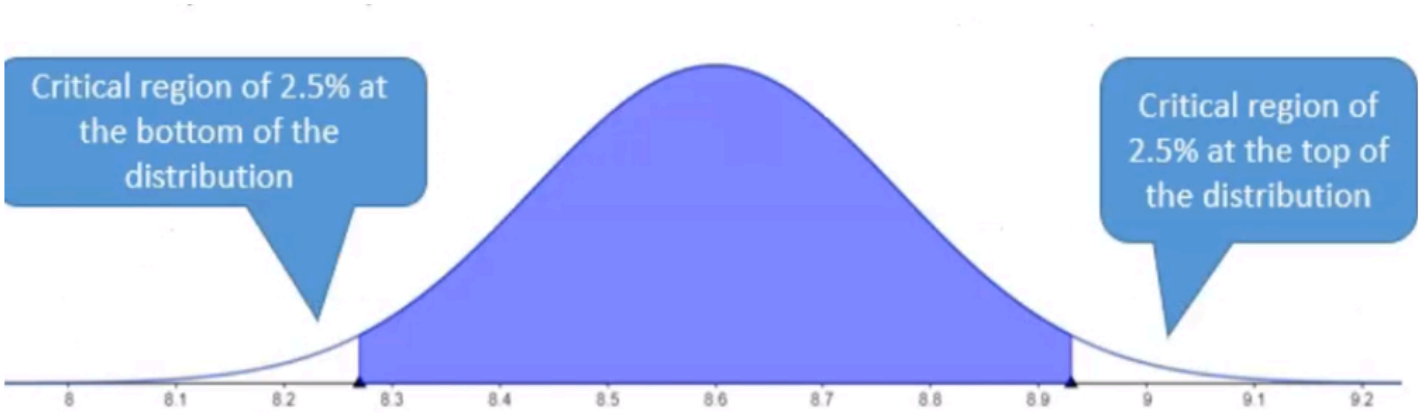
The 97.5th percentile is 8.93 cm. The upper rejection region is $\bar{X} > 8.93$. Our mean of 8.84 cm lies inside the acceptance interval.

Two tailed test example

Solution 75

Leaves of a rare plant species have normal lengths with mean 8.6 cm and standard deviation 1.2 cm. A botanist finds a clump of plants and measures 50 leaves. Their lengths total 442 cm. Test at the 5% level whether the mean differs from 8.6 cm. What should the botanist conclude?

$\bar{X} \sim N(8.6, 1.2^2/50)$.
$H_0 : \mu = 8.6$, $H_1 : \mu \neq 8.6$. The critical region has 2.5% in each tail.



Exercise 1

Solution 76

1. A random sample of size n is taken from a normal population with known standard deviation σ. Find the p-value and carry out each hypothesis test at the indicated significance level.

	σ	n	$\bar{x}$	H_0	H_1	Level
(i)	14	20	247	$\mu = 240$	$\mu \neq 240$	2%
(ii)	15	10	172.9	$\mu = 165$	$\mu > 165$	5%
(iii)	24	15	226	$\mu = 240$	$\mu < 240$	1%

(i)
$\bar{X} \sim N(240, 14^2/20)$. The upper-tail probability is 0.0127. The two-tailed p-value is $2(0.0127) = 0.0254 > 0.02$. Do not reject H_0; there is insufficient evidence that the mean differs from 240.

(ii)
$\bar{X} \sim N(165, 15^2/10)$. The one-tailed p-value is $P(\bar{X} > 172.9) = 0.0479 < 0.05$. Reject H_0; there is evidence that the mean exceeds 165.

(iii)
$\bar{X} \sim N(240, 24^2/15)$. The one-tailed p-value is $P(\bar{X} < 226) = 0.0119 > 0.01$. Do not reject H_0; there is insufficient evidence that the mean is below 240.

Exercise 2

Solution 77

Adult student masses are normal with mean 67.4 kg and standard deviation 3.8 kg. A sample of 24 has mean 65.8 kg. Assuming unchanged standard deviation, test at the 1% level whether the mean has decreased. Give the critical region.

2

$H_0: \mu = 67.4$, $H_1: \mu < 67.4$. Under H_0, $\bar{X} \sim N(67.4, 3.8^2/24)$.

The lower 1% cutoff is $67.4 - 2.3263(3.8/\sqrt{24}) \approx 65.60$ kg. Reject when $\bar{X} \le 65.60$.

The sample mean 65.8 kg is outside the critical region. There is insufficient evidence of a decrease.

Exercise 3

Solution 78

A machine produces ball bearings of intended diameter 15 mm. A random sample of 80 has mean 14.64 mm. Diameters are normal with known standard deviation 1.41 mm. Test at the 2% significance level whether the mean has changed, giving the critical region.

3

$H_0: \mu = 15$, $H_1: \mu \neq 15$. Under H_0, $\bar{X} \sim N(15, 1.41^2/80)$.

Put 1% in each tail. The limits are $15 \pm 2.3263(1.41/\sqrt{80})$, or 14.6333 and 15.3667 mm. Reject outside these limits. 14.64 mm falls between them. There is insufficient evidence of a change.

Textbook Examples

Solution 79

Fruit juice carton volumes are normally distributed with standard deviation 3 ml. The company claims a mean of 60 ml. An inspector suspects that it is overstated. A random sample of 16 cartons has mean 59.1 ml. State the hypotheses and test the complaint at the 5% level.

Null hypothesis $H_0 : \mu = 60$ ml.

Alternative hypothesis $H_1 : \mu < 60$ ml.

Variable Let X be the volume in an individual carton.

Population Under H_0, $X \sim N(60, 3^2)$.

Sample mean $\bar{X} \sim N(60, 3^2/16) = N(60, 0.75^2)$.

Test and conclusion

$P(\bar{X} < 59.1) = P(Z < -1.2) = 0.1151$.

$0.1151 > 0.05$, so do not reject H_0. The sample does not provide sufficient evidence that the mean carton volume is below 60 ml.

One carton and a sample mean X describes an individual carton. $\bar{X}$ describes the mean of a random sample of 16 cartons. Start from the distribution of X under H_0, then divide its variance by 16 to get the distribution of $\bar{X}$.

Finding the critical region

Solution 80

A machine produces bolts of normally distributed diameter, with mean 0.580 cm and standard deviation 0.015 cm. After servicing, a random sample of 50 bolts is taken. The standard deviation is unchanged.
(a) State the hypotheses and find the 1% critical region for a change in mean diameter.
(b) The sample mean is 0.587 cm. Comment.

Hypotheses $H_0 : \mu = 0.580$, $H_1 : \mu \neq 0.580$.

Population Under H_0, $D \sim N(0.580, 0.015^2)$.

Sample mean $\bar{D} \sim N(0.580, 0.015^2/50)$.

Critical values $z = \pm 2.5758$.

Critical region

$0.580 \pm 2.5758(0.015/\sqrt{50})$.

Reject for $\bar{D} \leq 0.574536\ldots$ or $\bar{D} \geq 0.585464\ldots$ cm.

Observation

$0.587 > 0.585464\ldots$. Reject H_0. There is evidence at the 1% level that the mean diameter has changed.

Two tails at 1% give 0.5% in each tail. Use the normal table or inverse normal on your calculator.

p	z	p	z
0.5000	0.0000	0.0500	1.6449
0.4000	0.2533	0.0250	1.9600
0.3000	0.5244	0.0100	2.3263
0.2000	0.8416	0.0050	2.5758
0.1500	1.0364	0.0010	3.0902
0.1000	1.2816	0.0005	3.2905

Standardise using the sample mean's standard error $\sigma/\sqrt{n}$.

Test Your Understanding

Solution 81

Edexcel S3 June 2011 Q7(a). Roastie's Coffee packets claim to weigh 250 g. A supermarket manager suspects the actual mean is lower. A random sample of 90 has mean 248 g and standard deviation 5.4 g.

(a) Test the manager's claim at the 5% level, stating the hypotheses. (5)

a

$H_0 : \mu = 250, H_1 : \mu < 250.$

For this large sample, $z = \frac{248-250}{5.4/\sqrt{90}} = -3.513\ldots$.

The lower 5% critical value is -1.6449. Reject H_0: the evidence supports the manager's claim.

Exercise 3G – Statistics Y2 pages 58-60

Solution 82

6. Bolt diameters D are normal with standard deviation 0.1 mm; 5% exceed 5.62 mm. Only diameters from 5.1 to 5.6 mm can be sold.

(a) Find the probability a bolt can be sold. (5)

(b) Of 12 bolts, find the probability fewer than three cannot be sold. (2)

(c) A second machine makes normally distributed diameters with standard deviation 0.08 mm. The mean of 20 is 5.52 mm. Test whether the population mean is less than 5.7 mm, at 2.5%. State the hypotheses. (4)

a

$(5.62 - \mu)/0.1 = 1.64485\ldots$, so $\mu = 5.4555\ldots$.

Then $P(5.1 \le D \le 5.6) = 0.9256$.

b

$N \sim B(12, 1 - 0.92558\ldots)$.

$P(N < 3) = P(N \le 2) = 0.9455$.

c

$H_0 : \mu = 5.7, H_1 : \mu < 5.7$.

Under H_0, $\bar{Y} \sim N(5.7, 0.08^2/20)$.

$z = (5.52 - 5.7)/(0.08/\sqrt{20}) = -10.06$. The p-value is below 0.025. Reject H_0: there is evidence that the mean is below 5.7 mm.

Exercise 3G – Statistics Y2 pages 58-60

Solution 83

7. European water vole masses M are normal with standard deviation 12 g. 2.5% exceed 160 g.
(a) Find the mean. (3)
(b) Of eight random voles, find the probability at least four exceed 150 g. (3)
(c) Water rat masses are normal with standard deviation 85 g. For a random sample of 15 the mean is 875 g. Test whether the population mean differs from 860 g at 10%. State the hypotheses. (4)

a

$(160 - \mu)/12 = 1.95996$, giving $\mu = 136.48$ g.

b

$p = P(M > 150) = 0.1299\ldots$. For $N \sim B(8, p)$, $P(N \geq 4) = 1 - P(N \leq 3) = 0.0129$.

c

$H_0 : \mu = 860$, $H_1 : \mu \neq 860$. Under H_0, $\bar{N} \sim N(860, 85^2/15)$.
The upper tail is 0.24715, above the 0.05 cutoff in one tail. Do not reject H_0: there is insufficient evidence that the mean differs from 860 g.

Exercise 3G – Statistics Y2 pages 58-60

Solution 84

8. Daily mean windspeeds are normal with standard deviation 3.1 knots. A random sample of 25 days at Heathrow in 2015 has mean 12.2 knots. Test at the 2.5% significance level whether the population mean exceeds 9.5 knots. State the hypotheses. (4)

8

$H_0 : \mu = 9.5$, $H_1 : \mu > 9.5$.
Under H_0, $\bar{X} \sim N(9.5, 3.1^2/25)$.
The upper 2.5% cutoff is $9.5 + 1.96(0.62) = 10.715\ldots$ knots. Since $12.2 > 10.715$, reject H_0.
There is evidence the population mean exceeds 9.5 knots.

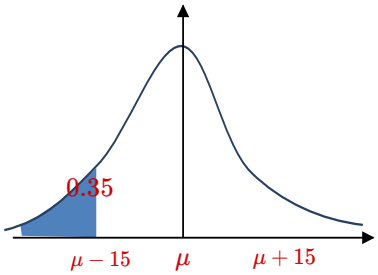
Conditional Probabilities

Solution 85

Edexcel S1 May 2014(R) Q4(c). The flight time from London to Malaga is normally distributed with mean 150 minutes and standard deviation 10 minutes. The flight time X minutes from London to another city is normally distributed with mean μ.

Given $P(X < \mu - 15) = 0.35$, find $P(X > \mu + 15 \mid X > \mu - 15)$. (3)

(c)



By symmetry $P(X > \mu + 15) = 0.35$, and $P(X > \mu - 15) = 0.65$.

$P(X > \mu + 15 \mid X > \mu - 15) = \frac{0.35}{0.65} = \frac{7}{13}$.

Every value above $\mu + 15$ is also above $\mu - 15$. The intersection is therefore just the event $X > \mu + 15$.

Test Your Understanding

Solution 86

Edexcel S1 Jan 2013 Q4(a,c). The time L hours that a phone lasts before it needs charging is normally distributed with mean 100 hours and standard deviation 15 hours.

(a) Find $P(L > 127)$. (3)

(b) Alice is about to make a six-hour journey. It has been 127 hours since she charged the phone. Find the probability it lasts to the end of her journey. (4; original part c)

a

$z = (127 - 100)/15 = 1.8$.

$P(L > 127) = 1 - \Phi(1.8) = 1 - 0.9641 = 0.0359$.

b

$P(L > 133 \mid L > 127) = \frac{P(L>133)}{P(L>127)}$.

$= \frac{P(Z>2.2)}{P(Z>1.8)} = \frac{1-0.9861}{1-0.9641}$

$= 0.3871\ldots \approx 0.39$.

IQ is distributed as $X \sim N\left(100, 15^2\right)$.
A person is considered to be 'pretty smart' if their IQ is at least 130.
Determine the median IQ of 'pretty smart' people.

Let a be the median IQ of 'pretty smart' people.

$P\left(X > a | X > 130\right) = 0.5$

$\frac{P(X>a)}{P(X>130)} = 0.5$

$P\left(X > a\right) = 0.5 \times P\left(X > 130\right) = 0.5 \times P\left(Z > 2\right) = 0.5 \times 0.0228$

$= 0.0114$

$2.277 = \frac{a-100}{15}$

$a = 134.155$

We could have obtained a directly at this point using the calculator's inverse normal distribution functionality.